# 1 Vectors in the plane

**Sheet 1** Thu 8 Oct. 2026 *Distance between two points and midpoint of a segment*

1. Warm-up · Quick Calculate $(-3)^2$, $(5-(-2))^2$ and $-3^2$.
2. Core · Method $A(1,2)$ and $B(4,6)$ in an orthonormal frame. Give the coordinates of $\overrightarrow{AB}$ and then the distance $AB$.
3. Core + · MCQ The midpoint of $[AB]$ with $A(-2,1)$ and $B(4,-3)$ has coordinates:
   **A.** $(1,-1)$ **B.** $(-1,1)$ **C.** $(2,-2)$ **D.** $(3,-2)$
4. Review · Quick Calculate $3x^2-2x+1$ for $x=-2$.
5. Thread · Quick Solve in $\mathbb{R}$: $3x+5=17$.

**Sheet 2** Mon 12 Oct. 2026 *Determinant of two vectors; collinearity criterion*

1. Warm-up · Quick Calculate $2\times(-3)-4\times 5$ and $(-1)\times 6-3\times(-2)$.
2. Core · Method Calculate $\det(\vec{u},\vec{v})$ for $\vec{u}\begin{pmatrix}2\\-3\end{pmatrix}$ and $\vec{v}\begin{pmatrix}-4\\6\end{pmatrix}$. What can be concluded?
3. Core + · True/false True or false? Justify. The vectors $\vec{u}\begin{pmatrix}1\\2\end{pmatrix}$ and $\vec{v}\begin{pmatrix}3\\5\end{pmatrix}$ are collinear.
4. Review · Construction Draw a circle with centre $O$ and two perpendicular diameters $[AC]$ and $[BD]$. What type of quadrilateral is $ABCD$? Justify.
5. Thread · Quick Programme: choose a number; multiply it by 4; add 5. a. What is the result for 3? b. What is the expression in terms of the starting number $x$?

**Sheet 3** Tue 13 Oct. 2026 *Collinearity and parallelism: the determinant method*

1. Warm-up · Quick Calculate the coordinates of $\overrightarrow{AB}$ for $A(1,-2)$ and $B(4,3)$.
2. Core · Method Are the points $A(1,2)$, $B(3,3)$ and $C(7,5)$ collinear?
3. Core + · Method For what value of $k$ are the vectors $\vec{u}\begin{pmatrix}k\\3\end{pmatrix}$ and $\vec{v}\begin{pmatrix}4\\6\end{pmatrix}$ collinear?
4. Review · Quick Calculate 20% of 350.
5. Thread · Translate A rectangle has length $x+3$ and width $x$ (in cm). Express its perimeter in terms of $x$ and simplify.

**Sheet 4** Thu 15 Oct. 2026 *Problem solving: vector, coordinate or geometric method*

1. Warm-up · Quick Complete: $ABCD$ is a parallelogram $\Leftrightarrow$ $\overrightarrow{AB}=\ldots$
2. Core · Method $A(0,0)$, $B(4,1)$, $C(5,4)$. Find $D$ so that $ABCD$ is a parallelogram.
3. Core + · Write-up $A(0,0)$, $B(3,1)$, $C(-1,3)$. Prove that $ABC$ is an isosceles right-angled triangle at $A$.
4. Review · Quick Write 450,000 in scientific notation.
5. Thread · Quick How many faces, vertices and edges does a cube have?